

SUMMARY TABLE: BIOLOGY GRADE 11

Sub-Strand 1.3: Taxonomy II — Teacher Reference (Pre-filled)

SUMMARY TABLE: BIOLOGY GRADE 11	
Sub-Strand	Sub-Strand 1.3: Taxonomy II
Driving Question	DRIVING QUESTION: How can we identify and classify living organisms accurately?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Why classify organisms?	Learners sorted a mixed set of Kenyan organisms (e.g. maize, beans, sukuma wiki, tilapia from Lake Victoria, a chicken, a cow, a butterfly, a mushroom) into groups using their own rules, such as 'edible vs not edible' or 'lives on land vs water'. Teacher note: allow WAIT TIME of at least 10 seconds after asking 'Why did the groups not agree?' and cold-call 3 groups to share their rule. Learners should notice that different rules produced different groupings, and that some organisms (e.g. a mushroom) did not fit neatly anywhere.	Organisms can be grouped by similarities and differences, but groupings made with different rules do not agree. Scientists therefore need a standard, agreed system based on observable and shared characteristics so that a farmer in Kisumu and a researcher in Nairobi mean the same organism when they name or group it. Classification makes the study of the huge diversity of life easier, shows evolutionary relationships and helps in identification and communication. Pedagogical note: link back to the anchoring phenomenon by asking 'Which rule would let us all agree?'	Model answer: Accurate classification needs a common set of rules that everyone follows, using features that can be observed and compared, not personal opinion such as 'useful' or 'tasty'. A standard system (taxonomy, with ranked groups from kingdom down to species) allows organisms to be identified and classified consistently. Look for learners using evidence from their own sorting activity (e.g. 'our groups disagreed because we used different rules'). Common misconception to address: classification by usefulness or habitat is not scientific classification.
Lesson 2 Kingdom Plantae — Bryophyta	Learners examined fresh or pictured mosses and liverworts collected from damp places (e.g. shaded rocks, stream banks in Kericho or the Aberdare forests) using hand lenses. They observed small size, thin leaf-like structures, hair-like rhizoids instead of true roots, and stalked capsules bearing spores. Teacher note: ask learners to sketch and label one specimen; circulate and ask 'What is missing that a maize plant has?' Wait at least 8 seconds before taking answers.	Division Bryophyta (mosses and liverworts) is identified by small size, rhizoids instead of true roots, no vascular tissue (no xylem or phloem), spores produced in capsules, and dependence on moist habitats because water is needed for fertilisation. Bryophytes absorb water directly through their surfaces. Pedagogical note: stress that lack of vascular tissue limits their height.	Model answer: Bryophytes stay small and live in damp, shaded places because they lack vascular tissue to transport water upwards and have only rhizoids for anchorage. They reproduce by spores in capsules, not seeds. So a moss can be classified accurately into Bryophyta by checking these features together, and this contributes to answering the driving question by showing that classification uses a set of shared observable characteristics. Misconception to watch: learners may think

			rhizoids are true roots; clarify that they lack vascular tissue and are mainly for anchorage.
Lesson 3 Pteridophyta and Spermatophyta	Learners compared a fern frond with sori (spore cases) underneath, a pine cone and a maize or bean plant with flowers and seeds, together with a moss. They recorded the presence or absence of vascular tissue, true roots, seeds, flowers and spores in a comparison table. Teacher note: cold-call 4 learners to fill in one row each and use WAIT TIME of 10 seconds before accepting corrections.	Pteridophytes (ferns) reproduce by spores and have vascular tissue and true roots but no seeds or flowers. Spermatophytes reproduce by seeds and have vascular tissue; they are divided into Gymnospermae (naked seeds, e.g. cypress, pine) and Angiospermae (seeds enclosed in fruits, flowering plants such as maize and beans), the latter further divided into monocots and dicots. Bryophytes lack vascular tissue. Pedagogical note: use maize (monocot) and beans (dicot) as familiar contrasts.	Model answer: Plants are separated into divisions by whether they have vascular tissue and how they reproduce. A fern is a pteridophyte because it has vascular tissue and spores but no seeds; a bean plant is a spermatophyte (angiosperm, dicot) because it has vascular tissue, flowers and seeds enclosed in pods. Using combined features allows accurate placement of any plant. Misconception: not all plants have flowers or seeds; ferns reproduce by spores.
Lesson 4 Using Dichotomous Keys	Learners used a simple printed dichotomous key to identify a set of local leaves or plants (e.g. maize, bean, sukuma wiki, grass, fern) and then wrote their own short key for four objects from the classroom or school compound. Teacher note: first model one couplet aloud, then let pairs work. Ask 'What happens if a couplet has more than two choices?' and wait at least 8 seconds.	A dichotomous key uses pairs of contrasting, observable characteristics in numbered couplets, so each step has only two choices and leads either to another couplet or to the identity of one organism. Good keys use clear features that can be seen (such as presence of wings, number of legs, leaf shape), not vague ones like 'big' or 'nice'. Pedagogical note: keys should avoid features that change with age or season.	Model answer: A dichotomous key makes identification accurate and repeatable because anyone following the same couplets reaches the same result. Each couplet removes organisms that do not match until only one remains. This helps answer the driving question by giving a step-by-step tool for identification when we do not know the name of an organism. Common errors: writing couplets with overlapping choices, or using subjective features; correct these during feedback.
Lesson 5 Kingdom Animalia — major phyla	Learners observed specimens or images of an earthworm, a snail, a grasshopper or crab, and a tilapia or chicken. They noted body segmentation, presence of a shell, jointed legs, exoskeleton and backbone. Teacher note: ask learners to record evidence in a table and cold-call 3 groups to justify one placement using a feature, not a guess.	Animals are classified into major phyla using observable body features: Annelida (segmented body, e.g. earthworms), Mollusca (soft unsegmented body, often with a shell, e.g. snails), Arthropoda (jointed legs and an exoskeleton, e.g. insects, crabs, spiders) and Chordata (a notochord or backbone, e.g. fish, birds, mammals). Pedagogical note: emphasise that Arthropoda is the largest phylum.	Model answer: A grasshopper belongs to Arthropoda because it has jointed legs and an exoskeleton; an earthworm belongs to Annelida because its body is divided into many similar segments; a tilapia from Lake Victoria belongs to Chordata because it has a backbone. Using structural features lets us place animals in phyla consistently. Misconception: learners may group by habitat (e.g. all water animals together); redirect to body features.
Lesson 6 Classes of Kingdom Animalia	Learners examined pictures or specimens of a tilapia, a frog, a lizard or chameleon, a chicken, a cow or goat, and also insects, spiders, crabs and millipedes. They recorded skin covering, limbs, breathing organs, reproduction and number of legs.	Phyla can be divided further into classes using more specific characteristics. Chordata is divided into Pisces (fins, gills, scales), Amphibia (moist skin, lives on land and in water, lay eggs in water), Reptilia (dry scaly skin, lay shelled eggs on land),	Model answer: Narrowing down from phylum to class needs more precise features. A chameleon is Reptilia because of dry scaly skin and shelled eggs; a frog is Amphibia because of moist skin and water-dependent breeding; a goat is Mammalia

	Teacher note: use WAIT TIME of 10 seconds after asking 'Which feature separates the frog from the lizard?'	Aves (feathers, beaks, wings) and Mammalia (hair or fur, mammary glands, most give birth to live young). Arthropoda is divided into Insecta (6 legs, 3 body parts), Arachnida (8 legs, 2 body parts), Crustacea (many jointed legs, 2 pairs of antennae) and Myriapoda (centipedes and millipedes, many segments). Pedagogical note: bats are mammals, not birds.	because it has hair and suckles its young. Classification is accurate when several features are used together. This adds detail to our model of how organisms are classified step by step from broad to specific groups.
Lesson 7 Dichotomous keys for animals and viruses	Learners used a dichotomous key to identify a set of local animals (e.g. housefly, spider, crab, earthworm, frog, tilapia) and then examined a table or diagrams of viruses (e.g. HIV, bacteriophage, tobacco mosaic virus) noting that they have no cell structure. Teacher note: let learners write one animal key for four animals and swap with a neighbour to test it; ask 'Where did the key fail?'	A dichotomous key uses paired contrasting observable features to identify organisms reliably. Viruses are non-cellular, have a protein coat (capsid) enclosing genetic material (DNA or RNA), and can reproduce only inside living host cells, so they are not placed in any of the kingdoms of living things and are often treated as borderline between living and non-living. Pedagogical note: connect viruses to diseases familiar to learners such as HIV and COVID-19.	Model answer: Keys work for animals and also for viruses when clear features such as shape, type of genetic material and host are used. Viruses are difficult to classify with the kingdom system because they are not cells and show life characteristics only inside a host. This shows that classification systems have limits and are revised as evidence grows. Misconception: viruses are not bacteria; they are far smaller and non-cellular.
Lesson 8 Synthesis and explanation	Learners reviewed their Driving Question Board, earlier tables and keys, and applied them to a new mixed set of Kenyan organisms (e.g. maize, fern, moss, pine, earthworm, snail, locust, tilapia, frog, chicken, goat), placing each correctly with evidence. Teacher note: ask learners to defend one placement in 3 sentences; cold-call at least 5 learners and allow 10 seconds of WAIT TIME before each response.	Plants are classified into Bryophyta, Pteridophyta and Spermatophyta (Gymnospermae and Angiospermae; monocots and dicots), and animals into phyla (Annelida, Mollusca, Arthropoda, Chordata) and then classes. Dichotomous keys help identify organisms using contrasting observable features, and viruses are non-cellular and fall outside the usual kingdoms. Pedagogical note: encourage learners to revise the class model they began in Lesson 1.	Model answer: We identify and classify living organisms accurately by observing structural and reproductive features, using a standard hierarchical system that moves from broad groups to specific ones, and using dichotomous keys with clear paired choices. For example, maize is a flowering plant (Spermatophyta, Angiospermae, monocot) and a tilapia is a chordate in class Pisces. Accept answers that cite specific evidence and mention that systems are revised with new evidence. Use this lesson to assess whether learners can transfer the method to unfamiliar organisms.

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